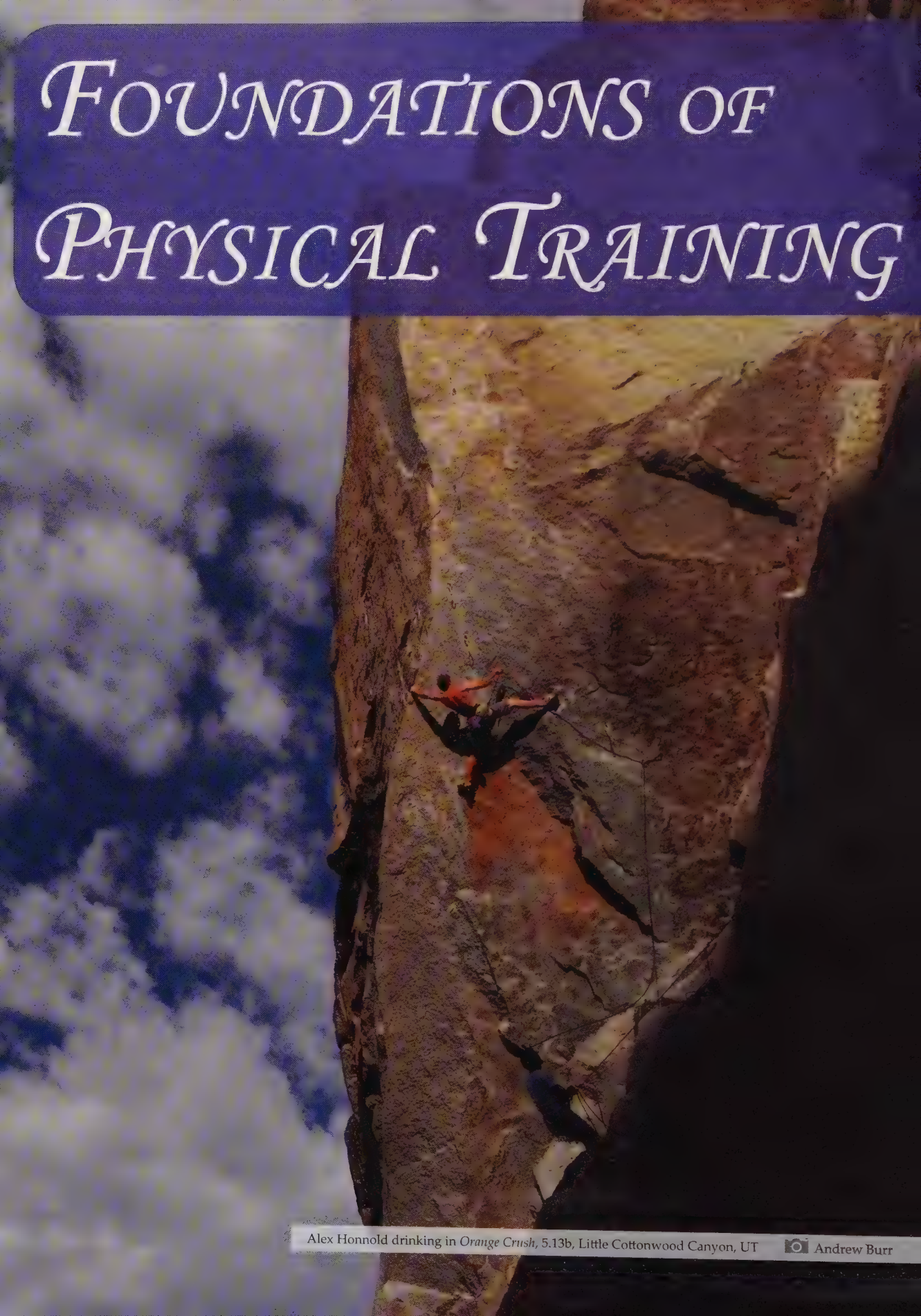


FOUNDATIONS OF PHYSICAL TRAINING



Alex Honnold drinking in *Orange Crush*, 5.13b, Little Cottonwood Canyon, UT

📷 Andrew Burr

CHAPTER

4

"I was so interested in the theory of training that I used to translate different articles about training from French and English as well. But of course at the beginning it was pretty instinctive. I tried to transfer training methods from other individual sports or programs to climbing."

— Josune Bereziartu¹²

INTRODUCTION

Ultimately, climbing is a physical act. Mental exercises like meditation and rehearsal, while important, will only take a climber so far. Eventually the hands and feet grasp stone, and the vertical dance begins. This dance requires strength to resist the pull of gravity, power to move dynamically, and endurance to overcome fatigue. Every climber on the path to continuous improvement will inevitably reach the point at which these physical attributes become a limiting factor.

Part II: Physical Training will discuss detailed, recommended approaches for improving each of these attributes, as well as the physiological basis for these recommendations. This chapter will explain basic training principles for athletes and how they can be applied to climbing training, as well as the framework for organizing the training methods that follow. The methods described will allow any climber to begin the process of achieving her potential.

PRINCIPLES OF EFFECTIVE PHYSICAL TRAINING

As detailed in **Chapter 1: Introduction**, *training* is defined as a systematic, disciplined, science-based approach to improving climbing performance. Athletes, coaches, and scientists have studied various training approaches for more than a century. Experience and experimentation have produced a number of principles that apply broadly to nearly every successful

Author's Note on Physiology

Each of the chapters in **Part II: Physical Training** includes extensive discussions on human physiology. It is our belief that a solid understanding of *why* a particular exercise is used can be very helpful to many athletes, because such knowledge builds confidence in the training program and encourages the athlete to commit to it. Just as a climber would be leery of committing to a long runout on bad rock if he thought he was off-route, it can be difficult to commit to a training program when you doubt its efficacy.

On the other hand, many athletes are willing to "buy in" without exhaustive explanations. These athletes may find the discussions on physiology tedious, excessive, or downright boring (while a small minority may even find them lacking in sufficient detail). It is not necessary to follow (or even read) the sections on muscle physiology. If you find yourself disinterested, feel free to use the "Quick-Start Guide to the Rock Prodigy Training Program" at the end of **Chapter 1: Introduction** to skip ahead to the juicy training details. The physiology will still be there waiting if you have questions down the road.

method of physical training. Training programs developed with these principles in mind stand the best chance of producing good results come performance time. Note that most of these principles apply to skill development as well as to physical training.

SPECIFICITY

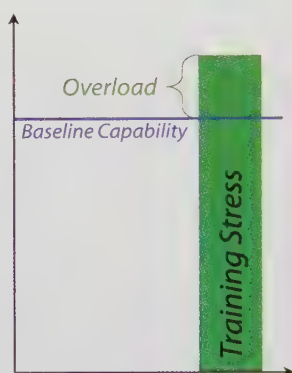
An athlete will primarily achieve physical adaptations in the systems that are stimulated by the training activity. For example, performing bicep curls with the left arm will increase strength in the left bicep, but not in the right bicep. This principle applies not only to the

FOUNDATIONS OF PHYSICAL TRAINING

physical location of stimulation in the body, but also to the *method* of training, including exercise intensity, duration, range of motion, and rest periods. For example, performing pull-ups will have a greater effect on the climber's ability to perform more pull-ups, and less effect on their ability to perform a lock-off (and vice versa). This sweeping principle will be referenced repeatedly in the chapters that follow, as it affects many aspects of training and is of vital importance to climbers. Climbing terrain offers infinite variation, giving climbers ample opportunity to train in ways that are NOT specific to their goals. Therefore, astute climbers will select training activities and terrain wisely to ensure the physical adaptations achieved are specific (or relevant) to their climbing aspirations.

OVERLOAD

Training stress must exceed the body's baseline capability in order to stimulate physical adaptation. In other words, training activities must exceed the athlete's "comfort zone." The body is prepared to handle a level of stress that is typical for the activities it routinely performs. It is necessary to exceed that typical level of routine stress to cause improvement. A climber who performs 10 body-weight pull-ups every day would need to increase this training stress (by increasing the number of pull-ups, increasing the resistance, or a combination of the two) to foster improvement in pull-up ability. Overload is difficult to achieve without the ability to *quantify* (or measure) the training load. This is a significant problem for climbers, since many climbing activities are extremely difficult to quantify accurately.



Overload is training stress that exceeds the athlete's baseline capability; it is essential to forcing adaptation.

RECOVERY

The act of training (particularly when overload is applied) actually makes the body weaker, causing *micro-trauma* to the muscles and connective tissues trained. Each training event must be followed by a period of rest, allowing the body to repair damaged tissues and replenish energy supplies (see figure in next column). The initial period of rest allows the body to return to its baseline capability. If sufficient training stimulus has been applied, additional rest will permit the body



Erin Drasler on Better Eat Your Wheaties, V8, Hueco Tanks, TX

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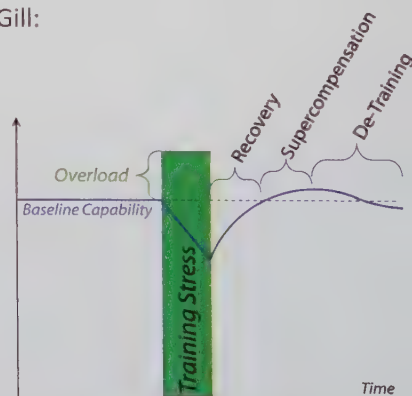
to *adapt*, achieving an improved baseline capability slightly above the previous baseline. This process is known as *super-compensation*. Climbers are notorious for climbing too much and resting too little. Heed the wise words of the legendary "father of modern bouldering" John Gill:

"Be sure to have frequent rest days, and don't overstress those small tendons. Gain your strength gradually and safely."

REVERSIBILITY

Physical adaptations achieved through training are *reversible*,

meaning they can be lost just as easily as they are gained. (Generally, the more difficult it is to achieve a physical adaptation, the longer it takes to lose it, and vice versa.) Once initial recovery and super-compensation periods are complete, further rest causes



Super-compensation as a result of Overload and Recovery. Note: diagram not to scale.

the body to weaken gradually. If an athlete becomes sedentary for an extended period of time, the body will revert back to untrained levels of fitness. This is known as *de-training*. While de-training should be avoided generally, there are some advantages. For example, the only way to convert “slow-twitch” muscle fibers to “fast-twitch” is through de-training, an important consideration for power athletes like boulderers (more on this to come).

REGULARITY

Training must be performed repeatedly at reasonable time intervals. While a single training session may cause some short-term improvements, those improvements will soon recede (see “Reversibility,” above). The ideal training frequency depends on many factors, including the training activity performed, the capacity of the athlete, and the desired adaptation. Low-intensity activities may be performed daily (or even two or three times a day in the case of elite endurance athletes), while higher-intensity activities (such as high-resistance weightlifting) may be performed as infrequently as once every three or four days. Climbers who wish to improve continuously must climb or train multiple times per week, at regular intervals, regardless of their goals or experience. *Simply climbing on the weekends is not sufficient.*

PROGRESSION

Training stress must gradually increase over the course of the training program. This principle is essentially the coordinated application of the overload and regularity principles (see figure below). Assuming adequate recovery, overloading the body will cause a slightly improved baseline capability, so training stress must be increased by a slight increment to overload the body during the next training event(s). Performing 10 pull-ups every other day will only train the body to maintain the ability to perform 10 pull-ups. On the other hand, regularly applied *progressive overload*

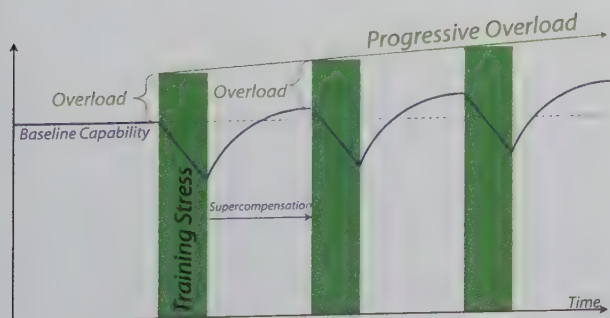
will gradually improve the body’s capabilities over the previous baseline level. This approach can be repeated many times over to achieve significant improvements over time. Climbing grades provide a convenient way to track this progression, so it may seem logical to simply attempt harder and harder routes as a means to force improvement. However, “grade chasing” is an inefficient way to pursue continuous improvement in the long run. For that, one needs a structured training program.

VARIATION

It would be nice if athletes merely had to apply the above principles indefinitely to achieve infinite improvement. Unfortunately, the body will eventually become resistant to the same, repeated training stimulus, resulting in the dreaded performance plateau. For this reason, athletes must vary training activities periodically in order to stimulate further physical adaptation. This principle has been distorted somewhat (by recent training fads touting “muscle confusion”) to suggest that athletes should never perform the same activity more than once. In reality, an athlete can progressively perform a given training activity for many consecutive weeks before improvement stagnates; in fact, doing so is the best way to ensure the principles of overload and progression are achieved. Furthermore, training should be varied to avoid mental fatigue (boredom) that ultimately results in declining motivation. Climbers often apply this intuitive principle by varying the style of climbing to follow the seasons (from bouldering to sport climbing, and to trad, multipitch, and ice climbing).

INDIVIDUALIZATION

Every person is different, so every body (and everybody) will respond differently to training. Therefore, a training program that works wonders for one athlete may be harmful to another. Each training program should be tailored to the individual athlete. Unfortunately, effective tailoring requires an observant and skilled coach or extensive personal experience. To apply this principle, most novice athletes should begin with a basic training program that applies broadly to many athletes (such as the Rock Prodigy program presented in the following chapters), and then make subtle changes over time as their experience and knowledge grow. The challenge of personalizing a *climbing* training plan is further complicated since goals (and the capabilities needed to achieve them) can vary dramatically between climbers, or even between seasons for the same climber. Even identical twins may need



Progression as a result of correct application of Overload, Recovery, and Regularity. Note: diagram not to scale.

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to individualize their training routines significantly if they are pursuing different climbing goals. As experience and ability grow, it will eventually be incumbent upon each climber to tailor the program provided in the following chapters to his or her specific goals.

TRANSFER

Train in a manner that replicates the conditions of performance to the greatest extent possible. This principle is similar to specificity, which is particular to physical adaptations within the body. Transfer, on the other hand, refers primarily to external performance factors such as time of day, environmental conditions, audience, etc. For example, climbers should train at the same temperature that they intend to perform at. While these factors may seem trivial, they are not. For example, studies have demonstrated significant improvements when the performance occurs at the same time of day as the preceding training sessions⁴. Often climbers have little control over the external conditions surrounding an important performance, so they should instead adjust the training conditions to ensure the best transfer.

CONTRIBUTING FACTORS

In addition to the nine general Principles of Effective Physical Training, the following factors contribute significantly to the successful execution of any training program:

QUANTIFICATION

The ability to measure or *quantify* training stress is critical to the principles of overload and progression. It is extremely difficult to ensure that the applied training stress exceeds baseline capabilities if the baseline capability cannot be defined, or the increment of *additional* stress (to achieve overload) cannot be measured. The same problem applies to the principle of progression. Often climbers attempt to overcome this problem through “feel” and intuition, but these methods are too easily sabotaged by the mind. An improved approach is to utilize training tools that allow climbers to precisely quantify (and therefore control) the training stress applied.

DOCUMENTATION

Related to quantification, collecting data on one’s training and performance activities, recording that data, and analyzing it for future use is essential to the principles of overload and progression, as well as to recovery, regularity, variation, individualization, and transfer. While developing an initial training plan

can be overwhelming, once it’s documented, creating future plans becomes much easier. Climbers who record their response to specific exercises can easily make slight adjustments to ensuing training plans, to approach ideal training exercises, frequencies, and loads. The training log sheets provided in the back of the book should simplify this process.

INTENSITY

Two primary parameters are used to measure training stress: *intensity* and *duration*. Intensity is the level of effort that must be applied to complete a given exercise. For some exercises, intensity is easily quantified in terms of the resistance used (for example, the load applied during a lifting exercise—weightlifters often track intensity as a percentage of “one-rep-maximum”). For other exercises, intensity cannot be easily quantified, but it can be qualitatively estimated. In other words, a subjective evaluation can be made of *how hard* an individual repetition *feels*, and that estimate of intensity can be documented. While this method is not perfect, each climber should at least be able to roughly define and track the intensity of a



In the case of weighted dead hangs, intensity is simply the quantity of weight applied (body weight, plus any weight added, or minus any weight removed).  Fred Marmsater

given training activity relative to other types of training, as well as the relative intensity of a given workout compared to past workouts of the same type. A good rule of thumb for climbers is, the harder one must pull on the holds in question, the more *intense* the movement. Training must be performed at the proper intensity in order to achieve the desired adaptation; more intensity is not always better! The following chapters will prescribe the proper intensity for the exercises presented.

DURATION

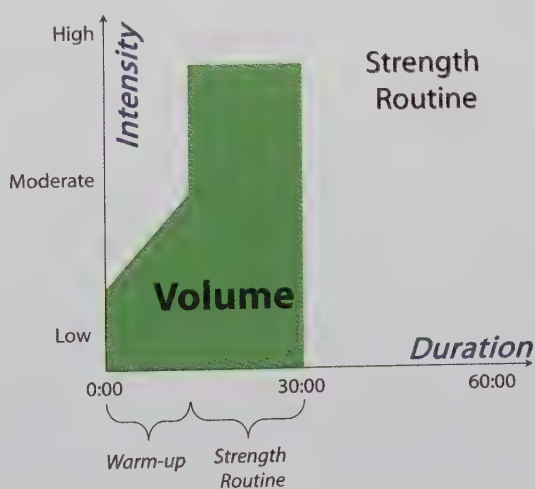
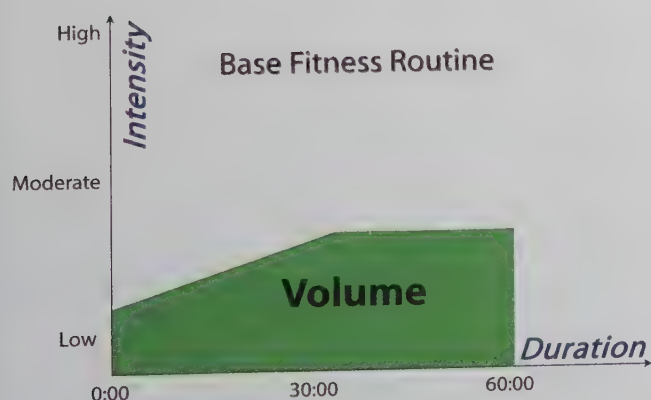
This is quite simply a quantitative measure of the time required to complete a training activity, and can apply to specific repetitions or to entire sets of repetitions. The principle of specificity suggests that climbers should pay special attention to the duration of exercises when training, in order to achieve the desired adaptations and meet their goals. Factors that will determine the specific adaptation to training include the total time required to complete a training activity, as well as the *Time-Under-Tension* (TUT), which is essentially the amount of time actually spent hanging from holds (for example, if a climber spends two minutes hanging from a jug rest, shaking hands in alternating fashion, each hand may only grasp the hold for 60 seconds or so TUT). In addition, since climbing performances are usually not time defined, tracking the number of movements (or vertical feet ascended) is also beneficial.

VOLUME

This measure is the product of intensity and duration. When tracking overload, progression, regularity, and recovery, *volume* is the most relevant metric. An increase in either the intensity *or* duration of a training activity will increase volume. Simultaneously increasing intensity *and* duration exponentially increases volume. In other words, doubling the intensity and duration results in *quadrupled* volume. Volume must be tracked carefully, and steadily increased to achieve appropriate progression throughout a training program.

ISOLATION

Muscles and their supporting structures can be trained most effectively when the principles of overload and progression are applied directly and individually to the component parts. This is because complex movements, requiring the use of many muscle groups, are limited by the weakest component required to complete the movement, so the exercise often ends well before the strongest muscles involved are overloaded. Isolation, on the other hand, allows the athlete to specifically target and improve each link (weak *and* strong). A relevant counterexample is the fingertip pull-up. Which muscles are trained to the point of overload? The latissimus dorsi, biceps, or finger flexors could be the weak link, and it is very unlikely that all three muscle groups reach overload at the same intensity and duration. Instead, it is better to isolate the finger flexors with “dead hangs” on the fingertip edge, isolate the lats with pull-ups on a bar, and isolate the biceps with dumbbell curls.



Volume is the “area under the curve” plotting intensity vs. duration. Two different examples are shown. The chart on the left describes a Base-Fitness Phase exercise, beginning with a low-intensity warm-up that gradually increases in intensity. Once the target training intensity is achieved, it is sustained while duration continues to increase. The chart on the right describes a Strength Phase exercise, beginning with a warm-up activity. While the strength routine lasts half as long, the total volume is identical to that of the much longer endurance routine.

Isolation vs. Complex Movements

Yet another in the unending parade of fitness fads is the preference for “complex movements” in strength training. The reasons for this are primarily that, first, complex movements are more specific to the way athletes perform in competition (running, jumping, etc.), and second, complex movements require the use of many muscle groups, thus creating high levels of “metabolic stress” (a key stimulator of muscle growth). However, much of this new obsession is fueled by the “bigger, faster, stronger” mentality valued by those training for team sports. Football players are keen to add mass in addition to strength, so anything that stimulates lots of muscle growth is championed.

Climbing, on the other hand, is all about strength-to-weight ratio. Remaining light is every bit as important as getting stronger. The ability to isolate specific muscle groups (those most critical to climbing performance) allows climbers to achieve overload precisely where it is desired, and limit growth where it may be harmful. As for metabolic stress, climbers easily achieve high metabolic stress where it is wanted—in the forearm muscles—without the use of complex movements. Also, the infinite variability of climbing terrain makes it essential that each link in the chain (including the strong links) is trained as effectively as possible. One move may involve a short reach from a tiny edge, where finger strength is the limiting factor. The next move may be a huge span from an enormous jug, where lock-off strength is emphasized. Training muscle groups individually ensures that each strength is at its best when it comes time to perform on the stone.

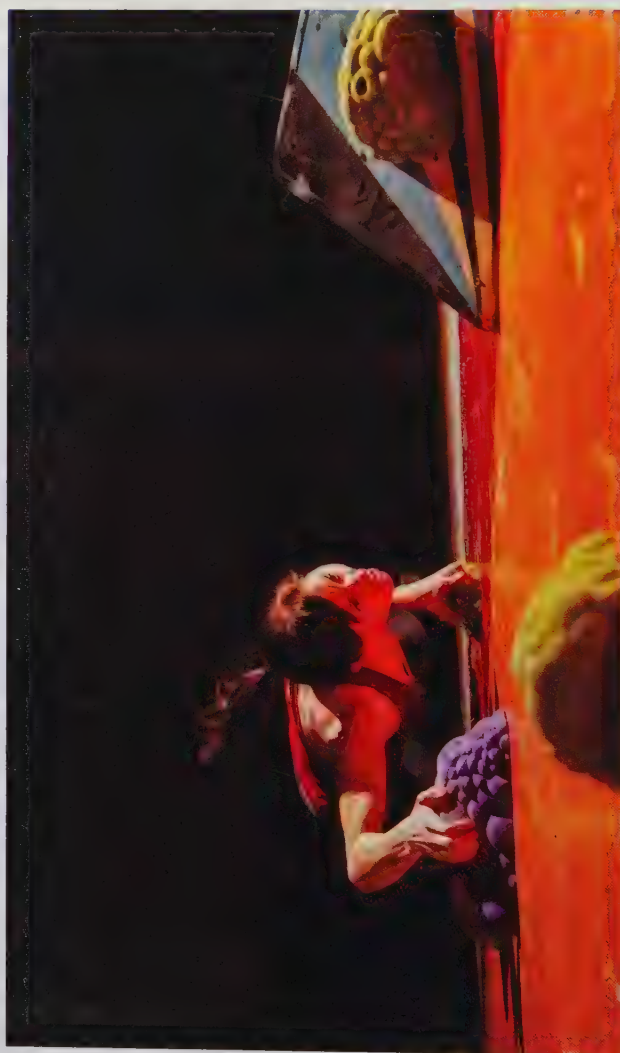
HOW TO TRAIN

Not everybody knows *how* to train. *It is one thing to follow a program with step-by-step instructions, but it is another to perform the activities in a manner that gets results. Even the best training program will fail if it is not implemented correctly.* Simply going through the motions, utilizing intensity levels that aren't challenging, or failing to face weaknesses is a waste of time. Owing to the principle of individualization, athletic training requires a certain amount of subjectivity. Each athlete must gauge for himself the relative intensity of his work output, and make adjustments to apply the desired training stress. Successful training requires constant excursions beyond the comfort zone, a good deal of temporary suffering, and large quantities of hard work.

“I think it's desire that gets results. People are often unwilling to acknowledge or give credit to hard work, seeming to prefer talent, even if it's wasted.” — Jerry Moffatt⁶

HARD WORK

Effective training requires hard work, and lots of it. This often comes in the form of dedication to a sometimes-monotonous schedule, perseverance in the face of physical discomfort, and discipline to follow through despite ephemeral inconveniences like an early wake-up time or unpleasant training conditions.



Hard work is an essential part of every effective training program.
📷 Dan Brayack

A penchant for hard work is not a character trait one is born with; it must instead be learned. Climbers with general athletic backgrounds are often at an advantage in this area, having learned how to work hard in a controlled environment, under the supervision of experienced motivators. A good coach, inspiring teammates, and healthy competition are good tools for developing a solid work ethic. These things are harder to come by in the sport of climbing, but the same basic approaches can be applied. Identify other climbers who perform at a high level and observe their work habits. While there are many paths to climbing success, they all involve hard work, and the best climbers all know how to work hard. Some may feign disinterest, but they all apply tremendous effort at the moment of truth.

Discomfort is a common barrier to hard work. Usually, hard work is unpleasant (in fact, any of the seven dwarfs still able to whistle could probably work a little bit harder). In addition to the obvious aversion to pain and suffering that tempts climbers to give less than their best, athletes are also encouraged to “listen to their body” or “back off at the first sign of injury.” These pieces of advice, while fundamental and neces-

sary, create a murky line that athletes are reluctant to cross for fear of injury or wasted effort. The solution to this conundrum is for each athlete to incrementally push ever so slightly beyond her comfort zone, monitor the effects, and then analyze the results. If all goes well, the athlete can push that much further beyond her previous limits the next time around. Though it may sound complicated, this is relatively straightforward to do with diligent application of the previously discussed training principles and contributing factors. This methodical approach to progressive overload is the essence of this training program.

“I can definitely be a work-aholic in certain ways. People who aren’t climbers a lot of times look at me and they’re like ‘you just play all the time for your job’ but I think if anybody were to really be able to experience exactly what I do, they’d be able to see that you have to get serious about it, and you have to be obsessive, and you have to really work hard.”

— Tommy Caldwell¹³

Too Hard or Not Hard Enough?

As discussed above, the intensity of a given training activity will determine how much effort must be applied, or how “hard” you must try to complete a repetition. You may already be performing the perfect training activity, but if the intensity is wrong, you won’t get the proper results. Effort levels vary greatly between climbers, and many just plain don’t know how much effort to apply, often because they don’t realize how much they are truly capable of.

Of course, there is potential for exercises to be too intense, especially when recovering from injury, but the opposite problem is far more common, particularly among those who have little experience with organized sports. If you’ve never really pushed your body to the limit, you will have trouble knowing just how much effort to put into a workout. Not all discomfort is bad, and it’s worth the sacrifice for each climber to discover the difference between good pain and bad. Here are some ways to help achieve the best possible training intensity:

- Prepare for important training days as you would for high-stakes performance days (like the last day of a road trip). Get proper sleep, visualize the activity in advance, and use a reliable routine to “get psyched” for the activity. This will get you in the proper mindset to apply the best possible effort to the training routine.
- Document the results of each workout, then use those records to create a “virtual training partner” in the future. Competition can bring out the best in everyone. Try to outdo your training partner each workout.
- Adjust your training intensity upward to the point that it becomes at least slightly unpleasant. Obviously, if every moment of every workout is miserable, the program will be unsustainable, but at key moments—like the last few repetitions of each exercise—things should feel uncomfortable. Sharp pains or chronic aches that persist many hours after exercise are not “good pain,” though feelings of nausea, slight dizziness, or fleeting muscle aches are symptoms of extraordinarily good effort, and should be sought from time to time.

Motivation and Effort

I went to school at the Air Force Academy, so I spent four years immersed in a process that takes people from across America and demands intense levels of hard work out of them. For those of us who had competitive sporting backgrounds, the physical training (PT) was tough but familiar. For those who didn't, the PT was *life changing*. They learned that the body is capable of much more than the mind wants to allow. While they certainly had exercised before, they always backed off at the onset of discomfort. Though this "breaking-in" process is often portrayed by Hollywood as cruel and demeaning, it isn't in real life (at least not these days). While yelling still occurs, the motivation is positive, based primarily in a strong sense of team (and competition with peer units), cultivating a desire to perform on behalf of your teammates and not let them down. This environment of intense competition and encouragement from comrades creates high arousal, and never-before-seen levels of effort. Many "average Joes" learn that they can put out a lot more than they realized. The military's generations-old method has awakened the potential and confidence of millions of soldiers over the years.

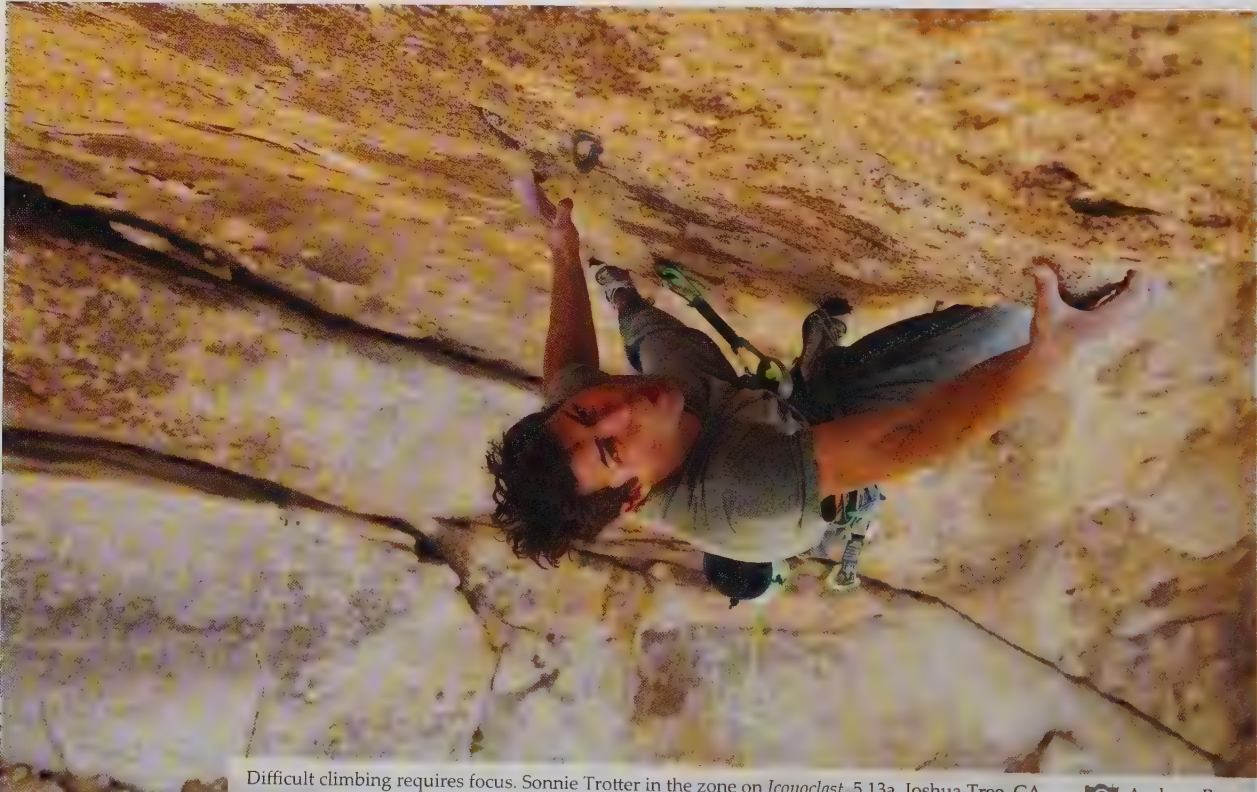
While these methods are impractical for every workout, once a person experiences his true capabilities a few times, he will be able to recall that level of effort at will. As climbers, we rely on inspiration and motivation to push past these mental barriers, but these may not be enough for an aspiring athlete with limited sports experience. If this sounds like you, and enlisting in the military isn't already on your "to-do" list, consider enrolling in a "boot camp"-style exercise class at a local health club, or try a few sessions of CrossFit to experience the power of external motivation.

— Mike Anderson

FOCUS

Mindlessly going through the motions of a training program will not produce the desired outcome. An athlete must be "present," living in the moment and keenly motivated to apply the best effort to every repetition, set, and exercise as often as possible. To do

this requires a high degree of focus, applied on a daily basis. It's often easy for climbers to "get psyched" for a hard onsight or redpoint attempt, because the rewards are tangible and close at hand. This same mentality must be applied to daily training activities to achieve the best results. Each day in the gym is an-



Difficult climbing requires focus. Sonnie Trotter in the zone on *Iconoclast*, 5.13a, Joshua Tree, CA.

© Andrew Burr

other opportunity to get better. What will be done with that opportunity? Will it be squandered in favor of a few moments of forgettable conversation about the weather?

With training cycles that last for months, often involving several weeks of training on plastic, maintaining this focus can be a significant challenge. Regardless, the effort and attention given to each workout (even those completed months before booting up below a project) have as much bearing on the eventual outcome as the effort put into the redpoint attempt.

Although the constant need to cultivate and sustain focus can be draining, it is a skill that improves with practice. Regularly executing this process can make it much easier to manifest that vital focus when it comes time to perform on the rock. Here are some strategies for applying proper focus to your day-to-day training activities:

- Eliminate external distractions by setting aside time dedicated solely to training. If possible, train in an isolated location away from distracting bystanders, and turn off any potentially distracting electronic devices.
- Select appropriate music, which can be helpful for achieving the proper state of arousal and intensity. Avoid radio, or other sources of noise that you can't control.
- Place photos of goal routes or inspirational messages nearby. If attention wanders unproductively, focus in on the photo of the goal route, visualize booting up at the base, and imagine the desires and anxieties that will accompany that process. Does that climber in your mind's eye believe he trained hard enough? Use the next exercise to ensure that the answer is "yes" when the actual redpoint day comes.
- Use controlled breathing. Begin breathing steadily and deeply before beginning the training activity, then maintain it throughout the activity. Practicing good breathing while training will make your breathing more reliable during performance.
- Utilize a consistent routine to prepare for training activities. This may include performing tasks (like booting up) in a consistent order, training at a certain time of day, or always listening to the same song during your warm-up.
- Keep your eyes focused on the task at hand. Vision can steer mental processes, so keep your eyes fixed on training tools, log sheets, timers, or other objects relevant to the training activity.

METHODS OF SYSTEMATIC TRAINING

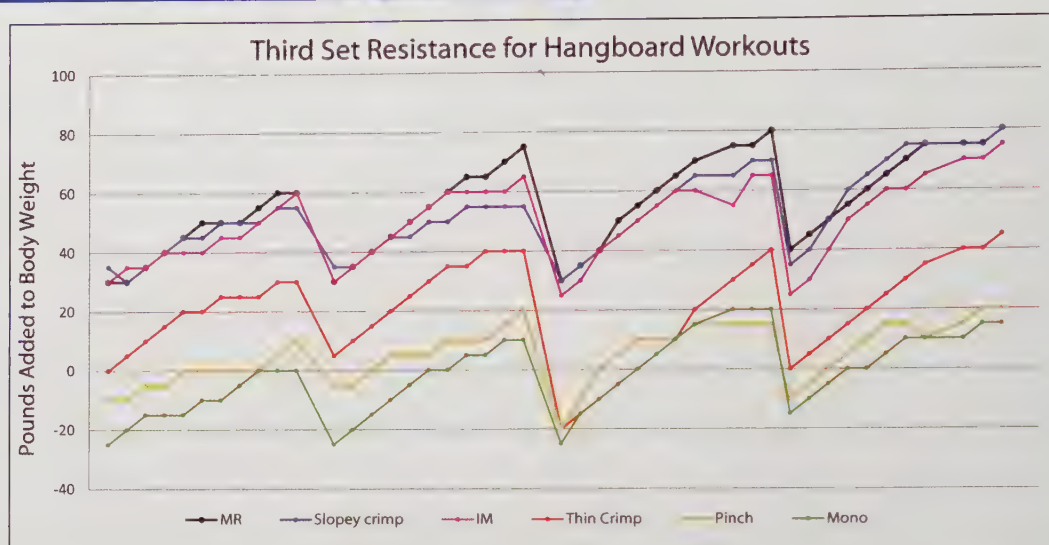
Athletes and coaches have designed many different methods of systematic physical training. This section will discuss some of these methods and their relative potential with respect to climbing training. All of these methods have their advantages, and an optimal climbing training program will borrow elements from each of these methods.

PROGRESSIVE RESISTANCE TRAINING (PRT)

Perhaps the oldest method of systematic training, PRT involves regularly performing a consistent set of exercises indefinitely, gradually increasing the volume (by increasing the number of repetitions and/or resistance) as the body adapts to training stress. The problem with PRT is that it lacks variability and the body eventually plateaus, failing to adapt or improve in response to further training stress. The majority of climbers unwittingly follow this approach, visiting the gym or crag on a regular basis and informally increasing training volume by climbing more pitches or attempting more-difficult routes (assuming they improve; however, improvements in technique and efficiency may result in reduced intensity and duration over time, thus causing regression). All successful athletes utilize this method to some extent, though usually as one element of a more comprehensive training program.

AUTO-REGULATORY PROGRESSIVE RESISTANCE TRAINING (APRT)

This new buzzword is a more deliberate method of "training to failure." Training to failure has been around for centuries, but its latest manifestation is now the hot fad in strength training. The concept is exceedingly simple: the athlete (or a coach) adjusts the training volume during the workout in order to increase the likelihood of achieving the desired amounts of intensity and duration. For example, if the athlete does not reach failure at the end of the pre-planned number of sets/repetitions, he would add more resistance and perform another set. If the athlete is unable to complete the prescription, he would reduce the resistance for the next workout. A recent study demonstrated substantially better results for this approach compared to classic PRT.⁵ This approach is highly recommended when performing the routines described in the following chapters.



The results of Auto-Regulatory PRT while training finger strength on a hangboard. Each color of line indicates a different grip position, and each dot indicates the amount of resistance used (for the given grip) during each workout. Note that the intensity is increased between workouts, resulting in significant strength gains over the season.

SPLIT TRAINING

Used by advanced athletes training with very high volumes, to spread that volume over multiple training days. For example, if a given training program demands performing 12 exercises (each with multiple sets), the athlete might perform the first four exercises on Day 1, the next four on Day 2, and the final four on Day 3. On Day 4 the cycle would repeat, with the athlete performing the first four exercises again. This approach works best when training diverse attributes, so that one attribute (such as a particular muscle group) can recover while the other is trained. Weightlifters often use this method by focusing on their upper body, core, and lower body in turn. Climbers can utilize this method in various ways, such as performing forearm and upper-arm training on Day 1, core-muscle training and Skill Development on Day 2, and Mental Training on Day 3.

LINEAR PERIODIZATION (LP)

As discussed in **Chapter 2: Goal-Setting and Planning**, periodization is a method that utilizes the basic concept of PRT while also varying the training activities to avoid training plateaus. Volume is steadily reduced over the course of a pre-defined season (usually around three to four months), while intensity is gradually increased, resulting in a performance peak at the end of the season. The season is sometimes referred to as a macro-cycle, and a given phase within the macro-cycle is known as a meso-cycle. This approach has many advantages,

not the least of which is that numerous studies agree it produces significantly superior results to other methods^{1,3,7,8,9,10,11}. Another significant advantage to climbers is that LP provides a framework for systematically training all of the many physical attributes (endurance, strength, power, skill) that contribute to climbing performance. A well-designed periodic training program can produce a synergistic culmination of these attributes, resulting in a towering performance peak lasting many weeks.

NON-LINEAR PERIODIZATION (NLP)

This method applies LP concepts to a shorter time period (usually one to two weeks) known as a micro-cycle. This method was developed for athletes whose sports demand near-peak performance throughout a long season, such as basketball players training for an 82-game season (versus those who train for many weeks to build up to a single event at the end of the season, like the Olympics). Each phase or type of training is performed within the micro-cycle, while volume and intensity are adjusted to create a mini-peak. For example, on Day 1 the athlete might train Base-Fitness, on Day 3: Strength, Day 6: Power, and Day 8: Power-Endurance, culminating in a performance on Day 10. Similarly, multiple types of training can be performed in a single day. The Rock Prodigy program borrows from this concept, beginning the macro-cycle with a classic LP approach (which is known to produce better long-term improvement than NLP), while incorporating a nonlinear approach during the Performance Phases.



Climbers with little control over their performance schedule, such as those training for a spread-out comp season, may benefit from Non-Linear Periodization. Daniel Woods on his way to the 2013 Sport Climbing National Title. Page Kuepper

THE ROCK PRODIGY TRAINING METHOD

As discussed in **Chapter 1: Introduction**, the Rock Prodigy training method combines various attributes

of the systematic training methods described above, while dutifully adhering to the Principles of Effective Physical Training. The intent is to maximize aspects of each method that are most useful to climbers. The basic framework of this program is a linearly periodic macro-cycle consisting of the following six phases:

- Base-Fitness (lasting two to four weeks)
- Strength (three to four weeks)
- Power (two to three weeks)
- Power-Endurance (two to four weeks)
- Performance (three to four weeks)
- Rest (one to two weeks)

The length of each phase is designed to promote adaptation continuously up to the verge of plateau, at which time the phase ends and a new phase begins. Phases can be varied to individualize the program in order to address specific goals or weaknesses. A typical macro-cycle will last 17 to 18 weeks,

SEP					OCT					NOV					DEC												
36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	1	2									
Novice Trainee:																											
Strength 3 wks					Power 2 wks					PE 3-4 wks					Sending 3-4 wks					Rest 2 wks							
Advanced Trainee:																											
Strength 4-5 wks					Power 3-4 wks					PE 3-4 wks					Sending 3-4 wks					Rest 1 wk							
Training																			Performance					Rest			

Example macro-cycles for novice and advanced climbers. Phase lengths can and should be manipulated depending on the climber's training experience, ability, and goals. Relatively novice climbers generally benefit more from skill-development training while improving strength and power relatively quickly, so increasing the length of the skill-intensive phases (Base-Fitness, Power-Endurance, Performance) and reducing the length of the Strength and Power phases would be appropriate. Advanced climbers generally need to spend more time training strength and power to force continued adaptation.

allowing for three “seasons” per year. Shorter seasons are also possible, but usually require reducing or eliminating one or more of the six Training Phases.

Within each Training Phase (or meso-cycle), progressive resistance training is used to force physical adaptation, and auto-regulation is encouraged to ensure overload is achieved in each exercise. Split Training is used to address the various attributes affecting climbing performance. During the Performance Phases (Power-Endurance and Performance), nonlinear periodization is incorporated by utilizing multiple types of training each day (for example, warming up with a skill-focused Base-Fitness exercise, performing strength/power bouldering exercises, and then finishing with a power-endurance-oriented linked bouldering circuit). This program has been utilized by many climbers over more than a decade, and the results are indisputable.

Each of the following chapters will describe an individual Training Phase in detail, finishing with an assortment of prescriptive routines to suit climbers with various levels of training experience. **Chapter 10: Building a Training Plan and Other Training Considerations** will present several detailed schedules, geared toward climbers of different abilities, describing precisely how to integrate all of these routines into a comprehensive periodic training program. Refer back to **Chapter 10** and **Chapter 2: Goal-Setting and Planning** as necessary to develop an individualized training schedule.

Finger Strength, First and Foremost

“Fingers, fingers, fingers. No world-class climber has weak fingers, but many can get away with weak bodies and weak shoulders. Make finger strength the number-one priority in your training, work on your weaknesses, and never drink on an empty stomach.” — Malcolm Smith¹⁴

The Rock Prodigy training method prioritizes training for finger strength and endurance over other physical attributes (such as core, pulling, or shoulder strength/power/endurance). Observing many climbers over many years, and at many crags around the world, has led us to the conclusion that finger strength and endurance are the primary physical limiters for most climbers. While the relative criticality of the fingers may vary depending on the climber’s preferred crags and/or climbing style, it is our observation that it is the rare climber indeed who couldn’t benefit from stronger fingers.

While the fingers are the primary *physical* limiter in climbing performance, their importance relative to technical and mental attributes may never be settled. For some moves, finger strength is absolutely everything; no amount of technique or tenacity will get you past an overhanging mono in an otherwise blank wall if you lack the finger strength. Other moves can be finessed, or overcome through strong will. Often the relative value of finger strength depends on the strengths, weaknesses, and goals of each individual climber. However, even when finger strength is not a limiting factor, improving finger strength has tremendous benefits over the course of your career.

Eventually (and quickly), every climber will progress to the point that the fundamental strength to pull on the holds becomes limiting. Furthermore, it takes many years of diligent effort to develop strong fingers, so it makes sense to begin addressing this inevitable weakness as early as possible. In the meantime, climbers can simultaneously develop the technical and mental elements of climbing performance (detailed training suggestions for these other attributes are included in **Chapter 3: Skill Development** and **Part III: Performing**). Wise climbers will endeavor to improve all of these areas, but finger strength is generally given preference in the text, as we believe most climbers will ultimately want to prioritize it in their own training. Other outstanding resources like *The Self-Coached Climber* and *The Rock Warrior’s Way* are better suited to comprehensively address the other elements of climbing performance.

SUMMARY

- The nine Principles of Effective Physical Training should be considered when designing any training program. These principles are: specificity, overload, recovery, reversibility, regularity, progression, variation, individualization, and transfer.
- Several contributing factors help enable effective training. These include: quantification, documentation, intensity, duration, volume, and isolation.
- Effective training requires hard work and focus. Both of these traits can be developed through practice and attention.
- There are numerous basic methods of physical training, including various forms of periodization and progressive resistance training.
- The Rock Prodigy training method incorporates elements of various time-tested training strategies to produce a synergistic performance peak at a predictable point in each climbing season.
- Physical, mental, and technical abilities all contribute significantly to climbing performance; however, the latter two have been covered exhaustively by other texts. This book focuses largely on finger strength because it has never been comprehensively described before yet is a primary determinant of climbing ability that all climbers could benefit from improving.



Certain moves demand brute finger strength. Sooner or later, every climber on the path of continuous improvement will run into such a move. Ryan Palo bearing down on *Disposable Hero*, 5.13b, Smith Rock, OR.  Peter Franzen